

A **process window** in photomask fabrication is the multidimensional envelope of process parameters—such as **e-beam write dose & focus, resist pre-bake/post-exposure bake (PEB) temperature, develop time, etch chemistry (Cl₂/O₂ ratio), plasma power, and chamber pressure**—within which the resulting pattern meets stringent specifications for:

- **Critical Dimension Uniformity (CDU)** (e.g., $\leq 1.0 \text{ nm } 3\sigma$ for advanced EUV masks),
- **Pattern placement accuracy,**
- **Sidewall profile (anisotropy, footing, undercut),**
- **Defect density** (bridges, pinholes, residues),
- **Line edge roughness (LER).**

This window is often visualized using **Focus-Exposure Matrix (FEM)** plots (for optical litho steps) or **Dose-Current/Defocus maps** (for e-beam writing). The **overlap region** where all specs are simultaneously satisfied defines the **usable operating space**.

A **wide, robust process window** enables high yield despite normal tool drift, environmental fluctuations, and material lot-to-lot variation—critical for cost-effective, scalable production.

Conversely, a **narrow process window** indicates high sensitivity to parameter shifts, demanding **extreme control** (e.g., $\pm 0.5^\circ\text{C}$ bake stability, sub-nm CD metrology feedback), increasing risk of scrap, rework, and cycle time delays.

Gradual degradation of consumable components—such as **O-rings, gas filters, chamber liners, showerheads, and resist dispense nozzles**—is a leading cause of **subtle process drift** in photomask fabrication. Even minor wear can:

- Introduce **metallic or particulate contamination** (e.g., from eroded O-rings or sputtered chamber walls),
- Alter **gas flow dynamics or plasma uniformity** (e.g., clogged filters or warped liners),
- Shift **etch rates or CD bias** due to changing surface chemistry or thermal conductivity.

To ensure long-term process stability and prevent yield excursions, implement a **data-driven, preventive replacement strategy** based on:

1. **Tool OEM recommendations** (baseline),

- 2. **In-situ sensor trends** (e.g., RF impedance, pressure decay, particle counts),
- 3. **Metrology correlation** (e.g., CD drift or defect spikes linked to consumable age),
- 4. **Failure Mode and Effects Analysis (FMEA)** for high-risk components.

Best Practice: Replace consumables **before end-of-life failure**—typically at **70–80% of their empirical lifespan**—to maintain the process within its validated window. Document all changes in the **equipment logbook** and trigger **requalification runs** (e.g., baseline CD checks) after critical replacements.

🔧 Photomask-Specific Examples			
Consumable	Failure Mode	Impact on Process	⬇
Chamber liner (etch)	Erosion → metal contamination	Increased pinhole defects, CD non-uniformity	
Gas filter (Cl ₂ /O ₂)	Clogging → flow instability	Etch rate drift, profile tilt	
O-rings (load lock)	Outgassing/cracking	Particle spikes, moisture ingress → resist scumming	
E-beam column aperture	Contamination buildup	Beam current instability → dose variation	

Abstract—In modern semiconductor manufacturing, one type of measured particle count data contains excess zeros, and the ratio of zeros in the measurements is usually larger than 50%. This type of particle count sample data cannot be well modeled by popular defect models such as Poisson, zero-inflated Poisson, generalized zero-inflated Poisson, and Neyman and Gamma-Poisson models. In this paper, a threshold-Poisson model was proposed to describe the particles with excess zero counts, and the method for parameter estimation was developed. Via comparison with those popular models by using 15 measured samples, it showed that the measurements are better modeled by the threshold-Poisson model. A control chart called threshold-c control chart was proposed and the control limits were derived. A reasonable minimum sample size for constructing control chart was also discussed based on simulations.

Index Terms—Control charts, excess zeros, particle, particle count, process control, quality control, statistical process control.

I. INTRODUCTION

AS THE chip density increases and semiconductor devices shrink in size, their yield, performance and reliability are sensitive to particulate contamination [1]. As a result, monitoring particles in semiconductor manufacturing is becoming more important. The particles in the air are measured and classified into discrete size using a particle counter, and in modern semiconductor manufacturing the particles can be measured by a laser particle counter. The commonly used method for monitoring the particle count is the c-chart [2]. The conventional c-chart is based on the assumption that the distribution of the number of defects is sufficiently well modeled by a Poisson distribution [2], [3]. In semiconductor manufacturing, we are usually concerned with the number of particles which are bigger than a setting value, causing the ratio of zeros in the

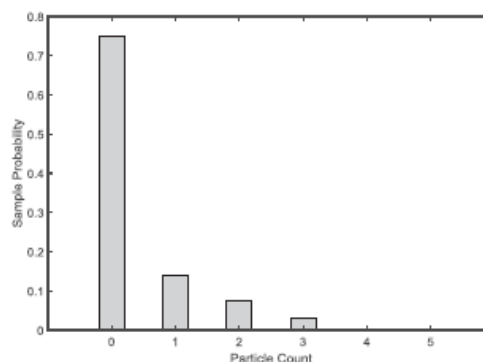


Fig. 1. An example of excess zeros in particle counts.

measurement to be usually larger than 50%. In this paper excess zero counts means the ratio of zeros is larger than 50%. Fig. 1 shows an example of the probabilities of the different particle counts. It illustrates that applying the traditional Poisson model is problematic and will lead to too many false alarms [4]–[7].

The distribution to describe the data with excess zero counts is defined as a zero-inflated distribution, and the most popular models are zero-inflated Poisson (ZIP) and generalized zero-inflated Poisson (GZIP). In addition, Neyman and Gamma-Poisson models are also used to describe the data with excess zero counts in recent years.

The ZIP model is proposed by Johnson and Kotz (1969). It assumes that the population consists of two types of measurements. The first type gives a Poisson-distributed count, which may be zero, whereas the second type always gives a zero count [4]. Xie and Goh (1993) [5] developed the SPC method based on the ZIP model, and proposed the original idea of the GZIP model informally at the end of his article. Böhning (1998) [6] found that these data such as the traffic accident data, criminal data and cancer registry data can be well modeled by the ZIP model. Xie *et al.* (2001) [7] investigated the use of the ZIP model and used it in SPC. Ridout *et al.* (2001) [8] provided a score test for testing the ZIP model. He *et al.* (2012) [9] proposed the CUSUM control chart based on the ZIP model. He *et al.* (2014) [10] proposed a combination of two CUSUM charts to detect the increases in both parameters of the ZIP process.

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The GZIP model was proposed by Chen *et al.* (2008), which is an extension of the ZIP model. The GZIP model assumes that the population consists of $n + 1$ ($n \geq 2$) types of measurements. The first n types give n different Poisson-distributed counts, which may be zero, whereas the last type always gives a zero count. Based on the GZIP model the Shewhart-type, CUSUM and ranked probability control chart were proposed by Chen *et al.* (2008) [11].

Dobbie and Welsh (2001) [12] modeled the data with excess zero counts by using the Neyman model, Dzupire *et al.* (2018) [13] described rainfall data with excess zero counts by using the Gamma-Poisson model.

In our opinion, there are two situations that would result in excess zero-defect phenomena. One is for “real defects”, that is the obtained defect count is the real defect count. In a ‘near zero-defect’ environment, it is understood that the individual product units are usually perfect, but there exists a random mechanism called shocks, which can lead to non-conformities in the sample whenever they are activated. The shocks occur randomly with a low probability so that a large number of zero nonconformity samples are observed [5]. The ZIP and GZIP models are proposed based on this situation.

The other situation resulting in excess zero-defect phenomena is for “measured defects”, that is, the defect count is the number of defects whose parameters are bigger than or equal to a setting value while the defects whose parameters are less than the setting value are ignored, leading to excess zero-defect phenomena. The zero defect here does not mean no defect but no defect bigger than the setting value. For example, in semiconductor manufacturing, if we only count the particles greater than a setting value in the clean room, a large number of zero particle counts will be observed. Zero particle count here does not mean no particle but no particle greater than the setting value. In practice, it is found that this kind of particle counts with excess zero counts cannot be well modeled by the popular models including Poisson, ZIP, GZIP, Neyman and Gamma-Poisson models.

In this paper, a model called threshold-Poisson model is proposed to describe this type of excess zero-defect phenomena, and the method for parameter estimation is developed. Via comparison with some other popular models by using 15 groups of measurements, it illustrates that the actual data is better modeled by this model. The control limits for the corresponding control chart named as threshold-c control chart are derived. The reasonable minimum sample size for constructing threshold-c control chart is discussed based on simulations.

The remaining parts of this paper are organized as follows: In Section II, we introduce the threshold-Poisson model, and develop the method for parameter estimation. Then the threshold-Poisson model is compared with some other popular models for defects. In Section III, the control limits for threshold-c control chart are derived, and some simulation results of sample size are given. In Section IV, we provide an illustrative example to explain the implementation of the threshold-c control chart. Finally, Section V summarizes the main work.

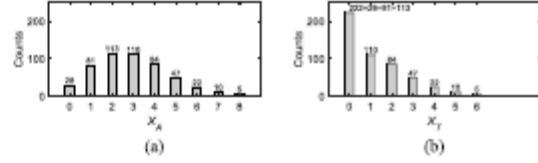


Fig. 2. An example illustrating the effect of measurement threshold.

II. THRESHOLD-POISSON MODEL FOR PARTICLES WITH EXCESS ZERO COUNTS

A. Threshold-Poisson Model

In semiconductor manufacturing the measured particle count is the number of the particles greater than a value set by a laser particle counter. This is equivalent to saying that there is a measurement threshold $T > 0$ in the particle counts. If the actual particle count X_A is bigger than the measurement threshold T , the particle count measured by particle counter will be $(X_A - T)$, and if not, the particle count measured by particle counter will be 0, leading to excess zero counts in the measured particle counts data. Therefore the measured particle counts X_T obtained from laser particle counter can be expressed as

$$X_T = \max(X_A - T, 0) \quad (1)$$

where X_A is the actual particle count, and X_T is the measured particle count obtained from a laser particle counter.

It is reasonable to assume that the actual particle count X_A follows the Poisson distribution, that is

$$p(X_A = m) = \frac{e^{-\lambda} \lambda^m}{m!} \quad m = 0, 1, 2, \dots \quad (2)$$

where $\lambda > 0$ is the parameter of the Poisson distribution.

According to Equations (1) and (2) the distribution of X_T can be expressed by Equation (3), which is defined as threshold-Poisson distribution in this paper.

$$p_m = p(X_T = m) = \begin{cases} \sum_{j=0}^{T-1} \frac{\lambda^j}{j!} e^{-\lambda}, & m = 0 \\ \frac{\lambda^{m+T}}{(m+T)!} e^{-\lambda}, & m = 1, 2, \dots \end{cases} \quad (3)$$

To illustrate the effect of the measurement threshold, a simple example is shown in Fig. 2. In this example, it is assumed that the measurement threshold $T = 2$, and the actual particle count X_A follows a Poisson distribution. Fig. 2 (a) shows the counts of actual particle count. Since $T = 2$, if X_A is 0, 1 or 2 then $X_T = 0$, so that the counts for $X_T = 0$ is $28 + 81 + 222 = 222$. If $X_A = 3$ then $X_T = 1$, so that the counts for $X_T = 1$ is 110. Similarly, the counts for $X_T = 2, 3, 4, 5$ and 6 correspond to the counts for $X_A = 4, 5, 6, 7$ and 8, respectively. Fig. 2 (b) shows the final result and the measurement threshold obviously results in too many zeros.

B. Parameter Estimation for Threshold-Poisson Model

It can be seen from Equation (3) that the theoretical probability of zero particles is

$$p_0 = \sum_{j=0}^{T-1} \frac{\lambda^j}{j!} e^{-\lambda}. \quad (4)$$

The cumulative distribution functions of the Poisson and the Chi-squared distributions are related in the following way [14]:

$$\sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda} = 1 - F_{\chi^2}(2\lambda; 2(T+1)) \quad (5)$$

where $F_{\chi^2}(2\lambda; 2(T+1))$ is the Chi-square cumulative distribution function at 2λ with $2(T+1)$ degrees of freedom. Since $p_0 = \sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda}$, then

$$\lambda = \frac{\chi_{p_0, 2(T+1)}^2}{2} \quad (6)$$

where $\chi_{p_0, 2(T+1)}^2$ is the upper p_0 percentage point of Chi-square distribution with $2(T+1)$ degrees of freedom.

In practice, we usually do not know λ and T , and they can be estimated from samples. Let k be the sample size, k_m be the counts of m in the sample, then the probability of m in the sample $p_{m, sample} = \frac{k_m}{k}$. Specially, the probability of 0 in the sample is $p_{0, sample}$. In case of excess zero counts, $p_{0, sample}$ is usually larger than 50%, and it is reasonable to replace p_0 with $p_{0, sample}$ in Equation (6) for parameter estimation, then

$$\lambda = \frac{\chi_{p_{0, sample}, 2(T+1)}^2}{2}. \quad (7)$$

There are two parameters in the threshold-Poisson model while there is only one mathematical relationship between λ and T , as shown in Equation (7). Therefore, a proper method for parameter estimation is proposed. The main idea of this method is to set the value of T , and then the corresponding value of λ can be obtained via Equation (7). After that the theoretical probabilities of each count value with T and λ can be calculated by Equation (3), and then the fit degree of theoretical probabilities and sample probabilities can be obtained. Setting different values of T , the corresponding λ and fit degrees can be obtained. The values of parameters T and λ with the best fit degree are selected as the estimated values of T and λ . In statistics, the determination coefficient, denoted by R^2 , provides a measure of how well observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model [15]–[17]. Therefore, we chose to use the determination coefficient to determine the best fit values of parameters T and λ . The theoretical determination coefficient is

$$R^2(T) = \lim_{M \rightarrow \infty} \left[1 - \frac{\sum_{m=0}^M (p_m - p_{m, sample})^2}{\sum_{m=0}^M (p_{m, sample} - \bar{p}_{sample})^2} \right] \quad (8)$$

where $\bar{p}_{sample} = \frac{1}{M+1} \sum_{m=0}^M p_{m, sample}$.

In practice, the theoretical determination coefficient is not easily obtained. After thorough study, it is found that:

(1) There is a minimum cut off point M which does not affect the estimates of T and λ , and also is smaller than the largest count value, so that setting M equal to the largest count value would not affect the estimates of T and λ .

(2) While M is equal to the largest count value, the $R^2(T)$ is only a little smaller than the theoretical value and the relative

deviation is smaller than 0.001. That is, when M is equal to the largest count value, the $R^2(T)$ has a very high accuracy.

By above discussion, it is reasonable and convenient to choose M equal to the largest count value of the samples, and then the Equation (8) can be simplified as

$$R^2(T) = 1 - \frac{\sum_{m=0}^M (p_m - p_{m, sample})^2}{\sum_{m=0}^M (p_{m, sample} - \bar{p}_{sample})^2} \quad (9)$$

where M is the largest count value of the sample.

We found that with T increasing, the determination coefficient increases to a maximum value and then decreases. Therefore, the steps of parameter estimation are as follows:

Step 1: Initial condition preparation.

Step 1.1: Obtain the largest count value M of the samples, and compute the probability of m in sample $p_{m, sample}$, $m = 0, 1, 2, \dots, M$.

Step 1.2: Set initial value of T , $T = 0$.

Step 2: Compute the determination coefficient $R^2(T)$;

Step 2.1: Compute the corresponding value of λ by substituting the value of T and the probability $p_{0, sample}$ into Equation (7).

Step 2.2: Compute the model probabilities p_m , ($m = 0, 1, 2, \dots, M$) by substituting the value of T and λ into Equation (3).

Step 2.3: Compute the determination coefficient $R^2(T)$ by substituting $p_{m, sample}$, p_m , and M into Equation (9).

Step 3: Similar to Step 2, the determination coefficient $R^2(T+1)$ can be obtained.

Step 4: If $R^2(T+1) > R^2(T)$, then let $T := T + 1$ and go to Step 2. Otherwise the values of T and λ are selected as the estimated parametric value.

In our work, the parameter estimation based on the principle of maximum likelihood estimation (MLE) is also studied, and the relationship between T and λ is

$$p_{0, sample} \frac{\sum_{j=0}^{T-1} \frac{\lambda^j}{j!}}{\sum_{j=0}^T \frac{\lambda^j}{j!}} + \frac{E(Y)}{\lambda} + \frac{T(1 - p_{0, sample})}{\lambda} - 1 = 0 \quad (10)$$

where $E(Y)$ is the average of the sample. If Equation (7) is replaced by Equation (10) for parameter estimation, with the similar steps of parameters estimation, the estimated value of T and λ could be obtained, then the corresponding R^2 can be also obtained. The simulation result shows that the MLE method for parameter estimation is not as good as the above method, because the average of R^2 by MLE is a little smaller and the standard deviation of R^2 is a little larger than that by the above method. The MLE method will not be discussed in detail.

C. The Advantage of Threshold-Poisson Model in Practical Application

15 Lasair II-100 particle counters labeled as A1~A15 are placed in 15 different positions of a semiconductor clean room to measure the particle count in the air, and the sample data are shown in the Table I.

The parameter estimation method proposed in Section II-B is applied to each sample data, and the results are shown in the last two columns of the Table I.

TABLE I
SAMPLE DATA IN A REAL CLEAN ROOM

Label	Sample size	k_n (Counts of n)	$P_{0, sample}$	T	λ
A-1	1178	$k_0=1067, k_1=73, k_2=26, k_3=11, k_4=1$	0.9058	4	2.3869
A-2	1176	$k_0=880, k_1=166, k_2=87, k_3=38, k_4=1, k_5=1, k_{10}=1, k_{11}=1$	0.7489	3	2.5403
A-3	1077	$k_0=793, k_1=129, k_2=84, k_3=63, k_4=3, k_5=2, k_6=2, k_{12}=1$	0.7363	5	4.3022
A-4	1074	$k_0=959, k_1=86, k_2=24, k_3=5$	0.8929	2	1.1367
A-5	684	$k_0=620, k_1=40, k_2=17, k_3=6, k_4=1$	0.9064	4	2.3815
A-6	684	$k_0=506, k_1=84, k_2=49, k_3=40, k_4=3, k_5=1, k_{10}=1$	0.7398	5	4.2813
A-7	670	$k_0=588, k_1=53, k_2=19, k_3=10$	0.8776	3	1.8829
A-8	668	$k_0=500, k_1=79, k_2=54, k_3=32, k_4=3$	0.7485	5	4.2283
A-9	303	$k_0=186, k_1=65, k_2=27, k_3=17, k_4=4, k_5=0, k_6=1, k_7=1, k_8=1, k_9=1$	0.6139	2	2.2331
A-10	242	$k_0=167, k_1=28, k_2=23, k_3=9, k_4=5, k_5=1, k_6=2, k_7=2, k_8=1, k_{10}=2, k_{13}=1, k_{21}=1$	0.6901	6	5.4742
A-11	239	$k_0=164, k_1=36, k_2=17, k_3=7, k_4=8, k_5=2, k_6=2, k_7=1, k_{11}=1, k_{30}=1$	0.6862	4	3.7053
A-12	235	$k_0=139, k_1=45, k_2=26, k_3=7, k_4=6, k_5=4, k_6=2, k_9=1, k_{12}=1, k_{17}=1, k_{19}=1, k_{20}=1, k_{40}=1$	0.5915	2	2.3172
A-13	219	$k_0=136, k_1=37, k_2=25, k_3=6, k_4=5, k_5=4, k_6=1, k_8=1, k_9=1, k_{10}=1, k_{13}=1, k_{19}=1$	0.6210	3	3.1172
A-14	213	$k_0=137, k_1=34, k_2=19, k_3=9, k_4=6, k_5=1, k_{10}=1, k_{11}=1, k_{12}=1, k_{13}=1, k_{26}=1, k_{27}=1, k_{51}=1$	0.6432	3	3.018
A-15	212	$k_0=140, k_1=37, k_2=14, k_3=5, k_4=6, k_5=1, k_6=2, k_8=3, k_9=2, k_{10}=1, k_{14}=1$	0.6604	2	2.0602

The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are widely used in the model selection. The smaller the AIC and BIC values are, the better the model is.

The AIC and BIC of the model are formally defined as [16], [17]

$$AIC = 2K - 2\ln(L) \quad (11)$$

$$BIC = K \ln(n) - 2\ln(L) \quad (12)$$

where K is the number of estimated parameters in the model, L is the likelihood function for the model, and n is the number of sample data. After parameter estimation, Equations (11) and (12) can be expressed as

$$AIC = 2K - 2k_i \sum_{i=1}^M \ln p_i \quad (13)$$

$$BIC = K \ln(n) - 2k_i \sum_{i=1}^M \ln p_i \quad (14)$$

where p_i is the modeled probability of count i of the corresponding model.

After calculation, the determination coefficient, AIC and BIC values of all models are shown in Table II, III and IV.

Table II suggests that: (1) The R^2 of the threshold-Poisson for each sample is equal or larger than that of other models. (2) The average of R^2 of threshold-Poisson is the largest and the standard deviation of R^2 is the smallest. (3) The smallest R^2 of threshold-Poisson is 0.9954, which means that all the particle data of A1-A15 can be well modeled by threshold-Poisson model.

From Table III and IV, it is observed that: (1) The AIC and BIC of threshold-Poisson model are smaller than those of the Poisson model. (2) The AIC and BIC of the GZIP($n=2$), GZIP($n=3$) and gamma-Poisson for A10-A15 are smaller than those of threshold-Poisson. This situation is caused by the abnormally large values. For example, if the maximums

of A10-A15 are deleted, then nearly half of the results will be reversed. (3) For about half of A1-A9, the AIC and BIC of the ZIP and Neyman are smaller than those of the threshold-Poisson. (4) The AIC and BIC of the threshold-Poisson are smaller than those of the popular defect models in most cases.

By above, the determination coefficient, AIC and BIC all suggest that the threshold-Poisson model has better performance for particles with excess zero counts than the other popular models.

Note that, the GZIP(n) means that the process contains n kinds of shocks and $2n$ parameters, and the parameter estimation method is given in [11].

Taking the data from 'A-3' and 'A-9' as examples to explicitly show that the threshold-Poisson model can fit the particle data with excess zeros well and stably, the results are shown in Fig. 3. Fig. 3 (a) shows that the threshold-Poisson, ZIP and Neyman modeled values are very close to the observed values of A-3 on the whole, while Fig. 3 (b) shows that the threshold-Poisson and Gamma-Poisson modeled values are very close to the observed values of A-9 on the whole, which means that the particle data with excess zeros can be well and stably modeled by the threshold-Poisson model. Fig. 3 also shows that it is difficult to model all the observed values accurately.

III. CONTROL CHART BASED ON THRESHOLD-POISSON MODEL (THRESHOLD-C CONTROL CHART)

A. Control Limits for Threshold-c Control Chart

In this paper, the corresponding control chart based on the threshold-Poisson model is named the threshold-c control chart.

The ratio of zeros of the particle data discussed in this paper is usually larger than 50%, so that the lower control limit LCL_T for the threshold-c control chart is always zero, that is, $LCL_T = 0$.

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-Poisson
A=1	1.0000	0.9923	1.0000	0.9981	0.9981	1.0000	0.9996
A=2	0.9997	0.9451	0.8159	0.9872	0.9872	0.9995	0.9974
A=3	0.9986	0.8842	0.9983	0.9915	0.9915	0.9983	0.9895
A=4	1.0000	0.9957	1.0000	0.9962	0.9962	1.0000	0.9998
A=5	1.0000	0.9908	1.0000	0.9984	0.9984	1.0000	0.9995
A=6	0.9986	0.8891	0.9982	0.9908	0.9908	0.9981	0.9901
A=7	0.9999	0.9860	0.9999	0.9972	0.9972	0.9998	0.9992
A=8	0.9989	0.8945	0.9975	0.9929	0.9929	0.9930	0.9826
A=9	0.9974	0.8899	0.9510	0.9557	0.9565	0.9789	0.9986
A=10	0.9986	0.6552	0.8997	0.9916	0.9916	0.9145	0.9947
A=11	0.9981	0.7304	0.8354	0.9839	0.9839	0.8445	0.9713
A=12	0.9970	0.5814	0.5813	0.9681	0.9681	0.5948	0.8909
A=13	0.9971	0.7075	0.8094	0.9759	0.9759	0.8434	0.9833
A=14	0.9987	0.4642	0.6962	0.9832	0.9832	0.7010	0.9087
A=15	0.9954	0.7187	0.8747	0.9738	0.9738	0.8987	0.9939
Average of R^2	0.9985	0.8217	0.8972	0.9856	0.9857	0.9176	0.9799
Standard deviation of R^2	0.0014	0.1677	0.1294	0.0124	0.0123	0.1254	0.0337

TABLE III
AIC OF DIFFERENT MODELS FOR REAL WORLD DATA

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-Poisson
A=1	940.52	1052.63	938.83	970.96	974.96	942.42	945.82
A=2	1980.93	2168.48	1995.28	2048.08	2052.08	1985.57	1990.55
A=3	1976.83	2265.15	1972.02	2008.55	2012.55	1966.83	1999.53
A=4	894.16	941.80	893.12	1035.22	1039.22	893.89	897.10
A=5	547.96	624.44	549.10	642.25	646.25	549.58	554.70
A=6	1239.67	1413.71	1235.53	1292.95	1296.95	1233.83	1252.97
A=7	652.22	722.35	653.51	759.09	763.09	652.49	657.63
A=8	1157.24	1284.82	1142.76	1227.66	1231.66	1154.29	1175.62
A=9	717.01	783.01	742.95	744.89	734.57	717.65	725.88
A=10	624.09	868.03	800.50	587.26	591.26	723.99	577.80
A=11	628.90	818.86	984.22	599.38	594.57	883.96	563.83
A=12	953.53	1142.07	1788.36	683.51	687.51	1579.48	698.63
A=13	634.90	781.83	794.85	610.48	614.48	708.31	574.23
A=14	938.87	1224.97	2178.30	602.96	606.96	1986.25	609.39
A=15	608.18	731.71	673.38	587.86	591.86	616.77	518.49

It is shown from Equation (1) that the upper control limit UCL_T for the threshold-c chart with parameters λ and T should be T less than the upper control limit UCL_0 for the traditional c chart with parameter λ . According to probability theory and statistics, the Poisson distribution is similar to the normal distribution only when λ is large enough, such as larger than 10. Table I suggests that λ is usually much smaller than 10 for particles in a clean room, which means that the corresponding Poisson distribution is far from the normal distribution. So, the UCL_0 for c charts need to be modified. Winterbottom (1993) proposed a modified 3σ upper control limit based on Cornish-Fisher expansion [18], [19], that is

$$UCL_0 = \lambda + 3\sqrt{\lambda} + 4/3. \quad (15)$$

The modified 3σ upper control limit UCL_0 is the (almost) exact 0.99865 upper control limit [18] [19].

Therefore, the upper control limit UCL_T for the threshold-c control chart should be

$$UCL_T = \lambda + 3\sqrt{\lambda} + 4/3 - T. \quad (16)$$

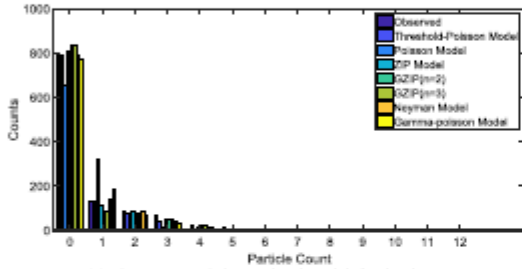
Then, the UCL_T for threshold-c control chart is also the (almost) exact 0.99865 upper control limit, so type I error is always around 0.00135.

B. The Minimum Sample Size Needed to Construct the Threshold-c Control Chart

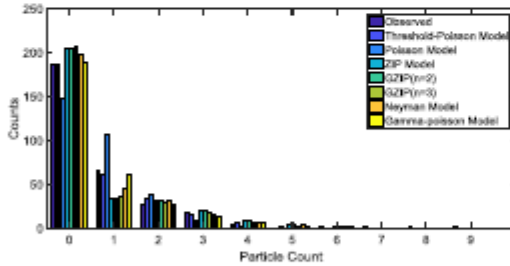
Shewhart control charts are usually constructed using 20~30 initial samples. The ratio of zeros of the particle data discussed in this paper is usually larger than 50%. Obviously, if the sample size is still 20~30, the inherent (common-cause) variation can have a significant impact on the sample probabilities of non-zero particle counts, which

TABLE IV
BIC OF DIFFERENT MODELS FOR REAL WORLD DATA

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-Poisson
A-1	950.67	1062.77	948.98	991.25	1005.39	952.57	955.96
A-2	1991.07	2178.62	2005.41	2068.35	2082.49	1995.71	2000.69
A-3	1986.79	2275.11	1981.98	2028.48	2042.44	1976.79	2009.49
A-4	904.12	951.76	903.07	1055.14	1069.10	903.85	907.06
A-5	557.02	633.49	558.16	660.37	673.42	558.64	563.75
A-6	1248.73	1422.77	1244.59	1311.06	1324.12	1242.89	1262.03
A-7	661.24	731.37	662.33	777.12	790.13	661.50	666.64
A-8	1166.25	1293.82	1151.77	1245.68	1258.69	1163.30	1184.63
A-9	724.44	790.44	750.38	759.75	756.85	725.08	733.30
A-10	631.07	875.01	807.48	601.22	612.19	730.96	584.78
A-11	635.85	825.81	991.17	613.29	615.43	890.91	570.78
A-12	960.45	1148.99	1795.28	697.35	708.27	1586.40	705.55
A-13	641.68	788.61	801.62	624.04	634.82	715.09	581.00
A-14	945.59	1231.70	2185.02	616.41	627.13	1992.98	616.11
A-15	614.89	738.42	680.10	601.28	612.00	623.48	525.20



(a) The contrast of observed and modeled value for 'A-3'



(b) The contrast of observed and modeled value for 'A-9'

Fig. 3. The contrast of observed and modeled value.

affects the calculating precision of UCL_T . Therefore, more initial samples are needed to construct the threshold-c control chart. In practice, a laser particle counter usually measures the particle count once per minute, and then 1,440 samples can be gathered one day. So, a reasonably large initial sample size such as several hundred initial samples is acceptable.

In order to determine the minimum initial sample size, lots of simulation experiments are carried out by the following steps:

Step 1: Set the values of p_0 and T , and compute the corresponding value of λ by Equation (7), then compute UCL_{TT} by Equation (16), where UCL_{TT} is the theoretical upper control limit.

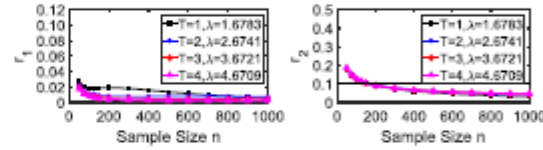


Fig. 4. Simulation results with $p_0 = 0.5$.

Step 2: Set the value of n , and generate 10^6 groups of the Poisson distribution random numbers with parameter λ , and each group contains n numbers. Then obtain the threshold-Poisson distribution random numbers by substituting the random numbers and T value into Equation (1).

Step 3: Estimate T and λ for each group of random numbers using the parameter estimation method proposed in Section II-B, and then 10^6 groups of estimated values of T and λ can be obtained.

Step 4: Compute the UCL_T for each group by substituting corresponding estimated values T and λ into Equation (16), and then we obtain 10^6 different UCL_T .

Step 5: Compute the mean and standard deviation of the 10^6 UCL_T , and they are represented as $\text{mean}(UCL_T)$ and $\text{std}(UCL_T)$. Compute the ratio $r_1 = (\text{mean}(UCL_T) - UCL_{TT})/UCL_{TT}$ and $r_2 = \text{std}(UCL_T)/UCL_{TT}$. That is, r_1 indicates the relative deviation between the expectation of UCL_T obtained by the parameter estimation method and the theoretical value UCL_{TT} ; and r_2 indicates the relative dispersion between the dispersion of UCL_T and the theoretical value UCL_{TT} .

Some simulation results are shown in Fig. 4~8.

(1) It is considered reasonable that if $r_2 \leq 0.1$, the dispersion of UCL_T is acceptable in engineering application. From Fig. 4 (a), Fig. 5 (a), Fig. 6 (a), Fig. 7 (a) and Fig. 8 (a), the minimum sample size needed to construct the threshold-c control chart (while $r_2 = 0.1$) for different p_0 can be obtained as shown in Table V. Table V suggests that the minimum sample size increases with the increase of p_0 .

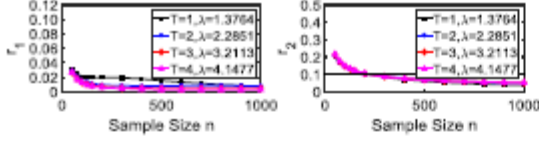


Fig. 5. Simulation results with $p_0 = 0.6$.

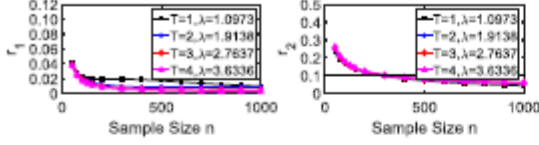


Fig. 6. Simulation results with $p_0 = 0.7$.

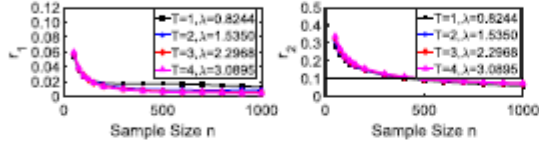


Fig. 7. Simulation results with $p_0 = 0.8$.

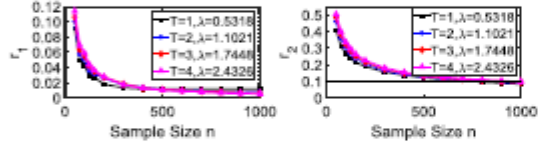


Fig. 8. Simulation results with $p_0 = 0.9$.

(2) Table V also shows the corresponding r_1 , and they are very close to 0, which means that the calculated UCL_T is almost unbiased.

(3) As seen from Table V, the non-zero number of 90 is enough for different p_0 .

In summary, to construct a threshold- c control chart, about 90 non-zero sample data needs to be collected, or the minimum expected sample size n_0 is about $90/(1 - p_0)$. Then in practice, the method of data collection for constructing the threshold- c control chart is: collect particle counts until about 90 non-zero particle counts are collected using a particle counter. It is easy to know that the sample size n obtained by this method follows a negative Binomial distribution. After study, it turns out that the probability of n falling inside the interval of $n_0[1 \pm (0.1 + 0.1p_0)]$ is about 95%. The maximum relative deviation is less than 20%, and it is acceptable in engineering application.

C. The Steps of Establishing Threshold- c Control Chart

Through the above analysis and discussions, the steps of constructing a threshold- c control chart are as follows:

Step 1: Collect particle counts until about 90 non-zero particle counts are collected using a particle counter.

Step 2: Estimate T and λ using the parameter estimation method proposed in Section II-B.

TABLE V
THE MINIMUM SAMPLE SIZE FOR DIFFERENT p_0

p_0	0.5	0.6	0.7	0.8	0.9
sample size	150	225	300	450	900
Counts of nonzero	75	90	90	90	90
r_1	≤ 0.02	≤ 0.02	≤ 0.02	≤ 0.02	≤ 0.015

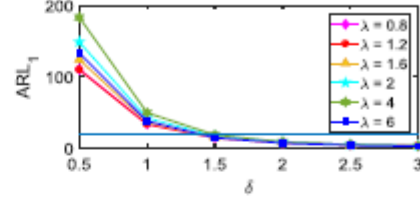


Fig. 9. The out-of-control average run length (ARL_1) for the proposed threshold- c control chart.

Step 3: Compute the UCL_T by substituting estimated value of T and λ into Equation (16).

Step 4: Draw the measured particle count and UCL_T in the threshold- c control chart.

D. The Performance for Threshold- c Control Chart

If λ shifts to another value of $\lambda + \delta\sqrt{\lambda}$, the probability of not detecting this shift on the first subsequent sample, or the type II error β , is

$$\begin{aligned}
 \beta &= p(X_T < UCL_T) \\
 &= p(X_T = 0) + p(X_T = 1) + \dots + p(X_T = \lfloor UCL_T \rfloor) \\
 &= \left[\sum_{j=0}^{\lfloor UCL_T \rfloor} p(X_A = j) \right] + p(X_A = 1 + T) + \dots \\
 &\quad + p(X_A = \lfloor UCL_T \rfloor + T) \\
 &= \sum_{j=0}^{\lfloor UCL_T \rfloor + T} \frac{(\lambda + \delta\sqrt{\lambda})^j}{j!} e^{-(\lambda + \delta\sqrt{\lambda})} \\
 &= \sum_{j=0}^{\lfloor UCL_T \rfloor} \frac{(\lambda + \delta\sqrt{\lambda})^j}{j!} e^{-(\lambda + \delta\sqrt{\lambda})} \quad (17)
 \end{aligned}$$

where $\lfloor x \rfloor$ means the nearest integer less than or equal to x . Then the average run length of the proposed threshold- c control chart when particle count is out of control (called ARL_1) is

$$ARL_1 = \frac{1}{1 - \beta}. \quad (18)$$

Fig. 9 presents a set of curves for ARL_1 , and it shows that the $ARL_1 < 60$ while $\delta = 1$ and $ARL_1 < 20$ while $\delta = 1.5$. The particle counter can collect the particle count once a minute, so that it is acceptable in practice. To detect the small shift more quickly, the corresponding CUSUM or EWMA control charts based on threshold-Poisson model should be used.

E. The CUSUM and EWMA Charts

The threshold- c chart is a Shewhart control chart, and the major disadvantage of the Shewhart control chart is

TABLE VI
SAMPLE DATA COLLECTED IN A CLEAN ROOM

NO	particle count	NO	particle count	NO	particle count	NO	particle count	NO	particle count
1	0	51	0	101	1	151	0	201	0
2	0	52	0	102	0	152	0	202	0
3	2	53	2	103	0	153	0	203	0
4	1	54	2	104	0	154	1	204	2
5	0	55	4	105	0	155	0	205	0
6	0	56	0	106	0	156	0	206	0
7	2	57	0	107	0	157	0	207	0
8	1	58	0	108	1	158	0	208	2
9	2	59	0	109	3	159	0	209	0
10	0	60	2	110	0	160	0	210	1
11	2	61	0	111	1	161	1	211	0
12	1	62	0	112	0	162	0	212	0
13	0	63	0	113	0	163	0	213	0
14	0	64	0	114	3	164	1	214	0
15	1	65	0	115	2	165	3	215	1
16	3	66	0	116	2	166	0	216	0
17	1	67	0	117	3	167	1	217	0
18	0	68	0	118	0	168	1	218	0
19	0	69	3	119	0	169	1	219	0
20	0	70	2	120	1	170	0	220	3
21	0	71	0	121	0	171	0	221	0
22	0	72	0	122	0	172	0	222	0
23	0	73	0	123	1	173	1	223	0
24	0	74	0	124	2	174	1	224	0
25	2	75	1	125	1	175	0	225	2
26	1	76	1	126	0	176	0	226	0
27	0	77	2	127	0	177	0	227	0
28	0	78	0	128	4	178	0	228	1
29	1	79	0	129	2	179	0	229	2
30	1	80	1	130	1	180	0	230	1
31	2	81	0	131	0	181	2	231	0
32	1	82	0	132	0	182	0	232	0
33	0	83	1	133	0	183	5	233	1
34	0	84	1	134	0	184	0	234	0
35	1	85	0	135	0	185	0	235	1
36	0	86	2	136	1	186	0	236	0
37	0	87	1	137	0	187	0	237	0
38	0	88	2	138	0	188	0	238	0
39	1	89	2	139	0	189	0	239	1
40	0	90	0	140	0	190	0	240	0
41	2	91	0	141	1	191	0	241	0
42	0	92	3	142	1	192	1	242	0
43	1	93	3	143	0	193	0	243	0
44	0	94	0	144	1	194	0	244	1
45	0	95	0	145	3	195	2	245	0
46	0	96	0	146	0	196	0	246	1
47	0	97	0	147	0	197	0	247	0
48	1	98	0	148	0	198	0	248	2
49	0	99	1	149	2	199	0	249	3
50	0	100	0	150	0	200	0	250	0

that it is relatively insensitive to small process shifts, such as, on the order of about 1.5σ or less [3]. Since the methods of establishing EWMA and CUSUM charts

based on the threshold-Poisson model are basically the same as those based on the other models, the method and key points of establishing EWMA and CUSUM charts

TABLE VII
STATISTICAL RESULTS OF THE SAMPLE DATA

Particle count(m)	0	1	2	3	4	5
Counts(k_m)	160	49	27	11	2	1
Sample probability ($p_{m, sample}$)	0.640	0.196	0.108	0.044	0.008	0.004

based on the threshold-Poisson model are only explained briefly.

1) *The CUSUM Chart Based on the Threshold-Poisson Model*: For the characteristic of the threshold-Poisson model, when the value set by the laser particle is not changed, the measurement threshold T will be constant. If the CUSUM is designed to detect a shift from λ to λ_1 , the CUSUM statistic B_i is updated according to the following

$$B_i = \max(0, B_{i-1} + K_i) \quad (19)$$

where $B_0 = 0$, and K_i is based on the log-likelihood ratio for a specified shift size, i.e.,

$$K_i = \begin{cases} \ln \frac{\sum_{j=0}^T \frac{\lambda_j^j}{j!} e^{-\lambda}}{\sum_{j=0}^T \frac{\lambda_1^j}{j!} e^{-\lambda_1}}, & X_{Ti} = 0 \\ \ln \frac{\sum_{j=0}^T \frac{\lambda_j^j}{j!} e^{-\lambda}}{\sum_{j=0}^T \frac{\lambda_1^j}{j!} e^{-\lambda_1}}, & X_{Ti} > 0 \end{cases} \quad (20)$$

where the X_{Ti} is the i th measurement.

The CUSUM chart signals when $B_i > h$, where h is the upper control limit, which can be obtained through simulation to achieve the desired in-control performance. And the approximate relationship between the h and the α error is $h = -\ln \alpha$ [11].

2) *The EWMA Chart Based on the Threshold-Poisson Model*: The EWMA statistic Z_i is updated according to the following

$$Z_i = rX_{Ti} + (1-r)Z_{i-1} \quad (21)$$

where $0 < r < 1$ is a constant and $Z_0 = \mu_0$, μ_0 is the theoretical mean with the parameters T and λ , and μ_0 can be obtained via the following equation (for more detail derivation see the Appendix)

$$\mu_0 = \lambda - T + \sum_{m=0}^T (T-m) \frac{\lambda^m}{m!} e^{-\lambda}. \quad (22)$$

The control limits for the EWMA control charts are as follows.

$$\begin{aligned} UCL &= \mu_0 + L\sigma \sqrt{\frac{r}{2-r} [1 - (1-r)^{2i}]} \\ LCL &= \mu_0 - L\sigma \sqrt{\frac{r}{2-r} [1 - (1-r)^{2i}]} \end{aligned} \quad (23)$$

where σ is the theoretical standard deviation with the parameters T and λ . And σ can be obtained via the following equation (for more detail derivation see the Appendix)

$$\sigma = \sqrt{\lambda - \sum_{m=0}^T (2\lambda - T - m)(T-m) \frac{\lambda^m}{m!} e^{-\lambda} - \left(\sum_{m=0}^T (T-m) \frac{\lambda^m}{m!} e^{-\lambda} \right)^2} \quad (24)$$

TABLE VIII
THE RESULTS OF ITERATIVE PROCESS

Threshold(T)	0	1	2	3
λ	0.4463	1.2630	2.1357	3.0323
$R^2(T)$	0.9619	0.9958	0.9996	0.9966

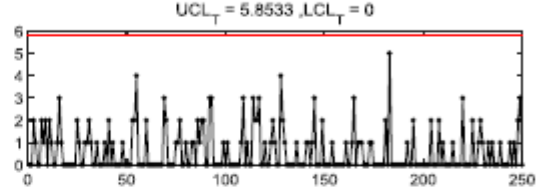


Fig. 10. An example using the threshold-c control chart to monitor particles with excess zero counts.

where L is the width of the control limits, and can be obtained through a Monte Carlo approach to achieve the desired in-control performance.

IV. AN APPLICATION IN REAL WORLD

According to Section III-C, an example is given to implement SPC (with the threshold-c chart) to monitor particles with excess zero counts in a clean room.

Step 1: In a clean room, we count the particles greater than a setting value using a Lasair II-100 particle counter, collect the particle count once a minute, and get 250 particle counts as initial sample data, which contains 90 non-zero particle counts. The sample data are shown in Table VI.

Step 2: Estimate T and λ using the parameter estimation method proposed in Section II-B.

From Table VI, the counts of different particles are shown in Table VII.

Then estimate T and λ using the parameter estimation method proposed in Section II-B, and the results of the iterative process are shown in Table VIII.

Table VIII suggests that the determination coefficient increases with the increase of T from 0 to 2, and then decreases with the further increase of T . And with the threshold $T=2$, the corresponding $\lambda = 2.1357$. The threshold-Poisson fits best with the observed measurement data, since $R^2 = 0.9996$ is very close to 1. So the parameter estimation results are $T = 2$ and $\lambda = 2.1357$.

Step 3: By substituting $T = 2$ and $\lambda = 2.1357$ in Equation (16), the upper control limits for the threshold-c control chart: $UCL_T = 5.8533$, can be obtained.

Step 4: After drawing the measured particle count and UCL_T in the threshold-c control chart, the threshold-c chart is obtained as shown in Fig. 10.

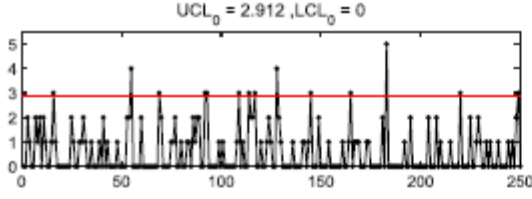


Fig. 11. Using c control chart to monitor the particle with excess zero counts.

Fig. 10 suggests that no point falls outside the UCL_T , which means that the variation of particle count in the air is in control.

As a contrast, if we choose the traditional c chart to monitor this process directly using the initial 250 samples, then we can get the c control chart as shown in Fig. 11.

Fig. 11 suggests that 14 points fall outside the UCL , which indicates that the process is seriously out of control. The result comes from misusing the c control chart.

V. CONCLUSION

In this paper, the statistical process control for monitoring particles with excess zero counts in semiconductor manufacturing is studied. The threshold-Poisson model is proposed and a suitable parameter-estimation method is developed. Then, by using 15 groups of measured particle data with excess zeros, the determination coefficient, AIC and BIC all suggest that threshold-Poisson model has better performance than the other popular models. After that, a control chart named the threshold-c control chart is proposed and the control limits are derived. Next the minimum sample size needed to construct the threshold-c control chart is discussed. The methods and key points for establishing the EWMA and CUSUM charts based on the threshold-Poisson model are also given. Finally, we use the threshold-c control chart to monitor the particle count, and the result is reasonable. Meanwhile, the determination coefficient is very close to 1, which illustrates the correctness of the model and the parameter estimation method. As a contrast, if we choose traditional c control chart to monitor the particle count, there would be too many false alarms.

APPENDIX

The derivation of Equations (22) and (24) based on the threshold-Poisson model

The theoretical mean μ_0 is deduced as follow:

$$\begin{aligned}\mu_0 &= \sum_{m=1}^{+\infty} m \frac{\lambda^{m+n}}{(m+T)!} e^{-\lambda} \\ &= \sum_{m=1}^{+\infty} (m+T) \frac{\lambda^{m+n}}{(m+T)!} e^{-\lambda} - T \sum_{m=1}^{+\infty} \frac{\lambda^{m+n}}{(m+T)!} e^{-\lambda} \\ &= \sum_{m=n+1}^{+\infty} m \frac{\lambda^m}{m!} e^{-\lambda} - T \sum_{m=n+1}^{+\infty} \frac{\lambda^m}{m!} e^{-\lambda}\end{aligned}$$

$$\begin{aligned}&= \left\{ \left[\sum_{m=T+1}^{+\infty} m \frac{\lambda^m}{m!} e^{-\lambda} + \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \right] - \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \right\} \\ &\quad - n \left(1 - \sum_{m=0}^T \frac{\lambda^m}{m!} e^{-\lambda} \right) \\ &= \lambda - T + T \sum_{m=0}^T \frac{\lambda^m}{m!} e^{-\lambda} - \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \\ &= \lambda - T + \sum_{m=0}^T (T-m) \frac{\lambda^m}{m!} e^{-\lambda}\end{aligned}$$

Similarly, the second order moment $E(X_T^2)$ can be obtained

$$E(X_T^2) = \lambda - \sum_{m=0}^T (2\lambda - T - m)(T - m) \frac{\lambda^m}{m!} e^{-\lambda}$$

Then the theoretical standard deviation σ is

$$\begin{aligned}\sigma &= \sqrt{E(X_T^2) - \mu_0^2} \\ &= \sqrt{\lambda - \sum_{m=0}^T (2\lambda - T - m)(T - m) \frac{\lambda^m}{m!} e^{-\lambda} - \left(\sum_{m=0}^T (T - m) \frac{\lambda^m}{m!} e^{-\lambda} \right)^2}\end{aligned}$$

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The fabrication of integrated circuits (IC's) has always promoted technical innovation. IC research and development, however, has traditionally been focused on improving the performance of the product, while paying relatively little attention to the efficiency of its production. Today, standard scientific manufacturing practices are finally finding their way into IC production. In most cases, the development of an industrial mentality coincides with the introduction of statistical process control (SPC). This paper undertakes a brief survey of standard SPC schemes, and illustrates them through examples taken from the semiconductor industry. These methods range from contamination control to the monitoring of continuous process parameters. Even as SPC is transforming IC production, the peculiarities of semiconductor manufacturing technology are transforming SPC. Therefore, the second part of this paper describes novel SPC applications which are now emerging in semiconductor production. These methods are being developed to monitor the short production runs that are characteristic of flexible manufacturing. Additional SPC techniques suitable for in situ multivariate sensor readings are also discussed.

I. INTRODUCTION

Historians divide the various evolutionary stages of modern manufacturing practice into six main periods [1], [2]. Throughout each of these periods, the objectives of efficiency and profitability have been pursued by focusing on different aspects of production. During the early 1800's, manufacturing practice was revolutionized by the introduction of machine tools that could achieve unprecedented mechanical accuracy. In the 1850's, the era of efficient mass production opened with the introduction of interchangeable parts. In the 1900's, Taylor introduced the concept of the scientific management of labor [3]. Around 1930, Walter Shewhart opened a new era by introducing statistical process control (SPC). The availability of computers made possible the introduction of the numerical control technologies that ushered us into the age of automation

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in the 1970's. All these advances finally culminated in the computer integrated manufacturing (CIM) systems of the 1980's.

While other industrial sectors had over two hundred years to absorb these changes, semiconductor manufacturing was thrust through them in less than two short decades. This transition has not been smooth and it is not yet complete. Nevertheless, powerful economic forces are finally transforming the laboratory art of IC fabrication into the modern science of manufacturing. Even though the cleanroom has traditionally been the domain of technology researchers and experimenters, the teachings of scientific manufacturing and the science of high volume production are finally being applied there. The subject of this paper is the application of SPC in modern semiconductor production.

SPC has been widely recognized as a tool that had technical as well as cultural impact on production [4]. Since its introduction in the 1930's by Walter Shewhart, SPC transformed the principles of production by transferring responsibility directly to the factory floor operator. This was accomplished by introducing a simple, yet powerful tool, the control chart. The control chart can be used as a gauge to detect and thus help eliminate unnecessary sources of variability. The inherent simplicity of these early control procedures empowered the operators with important decisions right at the point of production.

Since the 1930's the technical contributions to SPC have grown tremendously, in order to keep in pace with the advancing technology. Soon after its introduction, Shewhart's simple chart had to be augmented to accommodate various distributions, errors in measurements, small drifts as well as abrupt shifts, cyclic maintenance patterns, etc. These requirements led to the introduction of a large number of control procedures. Although some of the modern control techniques require substantial computational effort, relatively inexpensive computing hardware nonetheless enables the operator to use them at the point of production.

Today, the technology of production has evolved to the point, that new and complex SPC methods are necessary; in response, methods like multivariate statistics, time series modeling, intelligent control charts, etc., are gaining wider acceptance. These advantages are coming after a

host of statistical production techniques became popular in the semiconductor manufacturing industry during the late eighties [6]–[9].

The objective of this paper is to give a technical overview of the application and impact of SPC in semiconductor manufacturing. This paper consists of two major parts: the first is a short overview of the application of traditional SPC concepts, and the second is a summary of some of the modern SPC techniques that are finding use in semiconductor production.

Because of SPC's wide impact in the culture of production, it has been observed, and rightly so, that the introduction of SPC is as much a managerial challenge as it is a technical one. This paper will address the technical side of this issue.

II. SPC—BASIC CONCEPTS

SPC was introduced in the early 1930's by Walter Shewhart of Bell Telephone and Telegraph [9]. Shewhart's original objective was to provide a simple, intuitive way to summarize the history of the process. This summary was to serve as a reference to gauge present production performance. If future performance was found to be significantly different from its historical norm, then an alarm would be issued. By flagging significant process deviations and by finding and correcting their causes, the quality of a manufacturing process was improved.

More specifically, a process is said to be in *statistical control* when it displays nothing but the routine run by run variation. Unusual patterns and departures from that variation are indications that the process is *out of statistical control*. This implies that the process is experiencing a change that cannot be dismissed as routine variation. A central idea in SPC is that of the existence of an *assignable* (or "*special*") *cause* behind any significant deviation from the historical norm. In other words, an assignable cause is the reason behind every true alarm. The term "assignable" implies something that can be discovered and corrected, such as a chamber door that does not seal properly, a contaminated gas line, a miscalibrated film thickness meter, incoming wafers that are out of specifications, operator errors, etc. An assignable cause is to be contrasted to *chronic* or *routine* sources of variation, such as measurement errors caused by the inherent lack of precision in a photospectrometer, or the variation in the resistivity of a deposited layer, caused by the limited precision of the mass flow controllers in a furnace.

In general, *common causes* are the reason behind the imperfect run to run repeatability of a process. This repeatability is limited by the precision of the manufacturing equipment, the routine variation of the incoming material, the environmental cleanroom controls, etc. By definition, the operator cannot remove common causes from the process.¹

The role of an SPC procedure is then the formalization of the decision as to whether the process is operating under

statistical control. From a statistical point of view, SPC is a formal *hypothesis test*. This test will objectively choose between two hypotheses. The first hypothesis, known as the *null hypothesis* (H_0) is that the process is under statistical control. This assertion implies that there are no assignable causes of variation and that therefore the process is operating as consistently as possible. The second hypothesis, known as the *alternate hypothesis* (H_a) is that the process is out of statistical control. This implies that an assignable cause is present, and that this assignable cause should be discovered and removed in order to regain control of the process.

There are many ways that this hypothesis test is actually implemented. In its simplest form, the test consists of plotting one performance parameter against an upper and a lower control limit. These limits have been set in order to reflect the past behavior of the process. If these limits are met, then we accept H_0 and we declare that the process is under statistical control. If H_0 is rejected, then we adopt the alternate hypothesis H_a that stipulates the existence of an assignable cause. In this case, a misprocessing alarm is issued to the operator.

Naturally, it is essential that the alarms are reliably generated in the presence of the day to day variation which is characteristic of high volume production. Unfortunately, any statistical test which operates on a limited set of data is subject to errors. There are two types of errors that can be made while choosing among the two hypotheses. The first, known as a *type I error*, is to mistakenly reject H_0 , or, equivalently, to issue a false alarm. The second error, known as a *type II error*, is to mistakenly accept H_a , or, equivalently, to miss issuing an alarm. The probability of committing a type I error is also known as the *manufacturer's risk*, since it leads to unnecessary disruptions during production. The probability of a type II error is the *consumer's risk*, since it leads to producing defective products.

Once the hypothesis test has been defined, it is possible to estimate the probabilities of committing either one of these errors. This will permit the fine tuning of the statistical test so that production costs incurred by faulty decisions will be minimized. The following discussions address the calculation of the types I and II error rates in the context of the basic control chart.

A. The Control Chart and Examples of Its Application

In semiconductor manufacturing, more so than in other production areas, we often monitor a crucial process by means of measuring and recording one or more critical parameters. For example, the lithographic sequence can be monitored by, among other things, recording the thickness of the photoresist layer before the exposure. This layer is deposited by spinning a silicon wafer while the viscous photoresist solidifies. The target thickness of the photoresist is typically about 1 μm . Because of the mechanical nature of this operation, even if our equipment is functioning properly, the thickness of the layer on each wafer will vary. In addition, the measurement of the deposited layer,

¹Common causes can, however, be controlled by management actions, such as instituting tighter specifications on incoming material, retraining the operators, and upgrading critical pieces of equipment.

usually done with a photospectrometer, will be subject to calibration errors. The combination of these sources of variability give us a photoresist thickness that, as recorded, appears to be statistically distributed with a routine run to run variation. The goal of SPC is to implement a simple procedure that will reliably flag any significant deviation beyond the routine run to run variation.

The control chart is a simple and effective graphical representation of the process status. It can also serve as a very basic implementation of the hypothesis test discussed in the previous section. There are many types of control charts, each suitable to a different application. We will first describe the simple $\bar{X}$ chart², as a vehicle for illustrating some fundamental SPC concepts, such as the *types I and II errors*, the *average run length*, and the *operating characteristic function*.

The $\bar{X}$ chart is based on the assumption that, when the process is in control, the monitored variable is distributed according to a normal distribution with a known mean μ and a known sigma σ , symbolically:

$$x_i \sim N(\mu, \sigma^2). \quad (1)$$

Using the $\bar{X}$ chart consists of grouping and averaging n readings of x , defined as

$$\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i. \quad (2)$$

Under the assumption that each reading is *Independently and Identically Normally Distributed* (hereafter to be known as the *IIND assumption*), the arithmetic average can be shown to be distributed according to another known distribution given as

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right). \quad (3)$$

Pictorially, this control scheme is implemented by plotting the value of the group average versus time. If this value falls within a zone of high likelihood (determined by the known distribution of the plotted statistic), then we conclude that the process is under control. If the value falls outside this high likelihood region, the process is considered to be out of control. Traditionally, the high likelihood region is chosen to be within $\pm 3\sigma_{\bar{x}}$, where $\sigma_{\bar{x}}$ is the standard deviation of the arithmetic average. It can be shown that the *three-sigma* control limits yield a probability of type I error equal to 0.0027. These limits are defined as follows:

$$CL = \bar{x} \pm 3\frac{\sigma}{\sqrt{n}}. \quad (4)$$

If a point falls outside this zone, then we can conclude at an *0.0027 level of significance*, that this point is now generated by a different distribution, one that has shifted its mean, its variance or both.

²In this document, the symbol $\bar{x}$ indicates the arithmetic average of the random variable x . Most statistical symbols are not consistent across the literature. Whenever possible, I have adopted the symbols used in [9].

As an example, consider the $\bar{X}$ chart that was implemented to control the thickness of photoresist in the Berkeley microfabrication laboratory. Assuming that we know the mean and variance of this chart, we plot the average thickness versus time in Fig. 1.

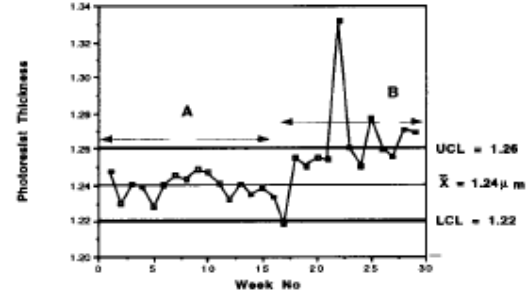


Fig. 1. $\bar{X}$ chart of photoresist thickness in the Berkeley Micro-fabrication Laboratory.

In addition to the types I and type II risks, the *average run length* (ARL) is also an important characteristic of the control chart. The ARL is defined as the average number of points plotted between alarms, false or otherwise. The ARL is a function of the process status and the type I and II risks. In Fig. 1, the process appears to be under control in the region designated as A. Here, we want the ARL to be as long as possible (because any generated alarms will be false), and in fact $ARL = 1/\alpha$. If $\alpha = 0.0027$ (for three sigma control limits), then the average number of samples between alarms is about 370.

Again in Fig. 1 the process appears to be out of control in the region designated as B. Here, we want the ARL to be as short as possible, and in fact the *out of control* ARL is equivalent to $1/(1 - \beta)$. In this region β , the type II risk, is related to the size and the type of the process shift, as illustrated in Fig. 2.

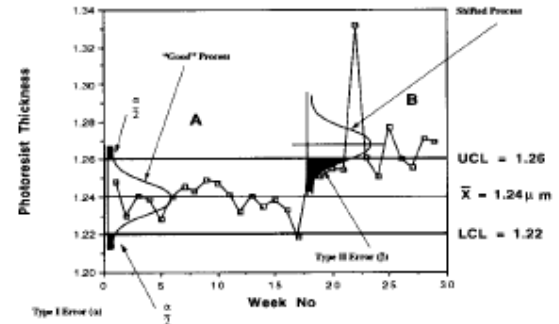


Fig. 2. Relations between type I and type II risks for an $\bar{X}$ chart.

Assuming that the process goes out of control because its mean has shifted while its variance stayed the same, it is possible to plot β versus k , the amount of the shift

(expressed in units of standard deviation) and the sample size n , which we employ in the calculation of the average. In Fig. 3, which has been adapted from [9], we show such a plot, known as the *operating characteristic function* of the chart. Next we discuss the application of some of the more sophisticated traditional SPC tools.

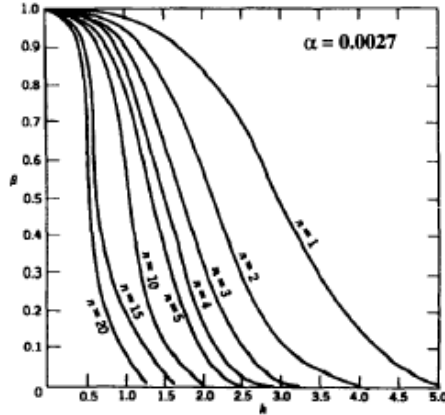


Fig. 3. The operating characteristic function of the $\bar{X}$ chart.

B. Controlling Location and Spread

In semiconductor manufacturing as well as in most manufacturing disciplines, production quality can be degraded by a shift of the mean or by an increase of the variance. For this reason, additional charts were created as companions to the $\bar{X}$ chart, in order to guard against an increase in the process spread. Two types of such charts are described below.

1) *The $\bar{X}$ -R Charts*: The simplest example of such a chart is the range (R) chart. The range of a group of measurements, defined as $R = x_{\max} - x_{\min}$, is a statistic with a known distribution [9]. This statistic is related to the standard deviation of the normal distribution that generated the measurements. More specifically, the average range $\bar{R}$ of m of groups can be shown to estimate the sigma of the distribution that generated the group of measurements:

$$\hat{\sigma} = \frac{\bar{R}}{d_2}. \quad (5)$$

Here d_2 is a proportionality factor which depends upon the size n of the group, and whose values have been tabulated in many standard textbooks of applied statistics. Furthermore, the standard deviation of the estimated range is also related to the standard deviation of the group:

$$\sigma_R = d_3 \sigma \quad (6)$$

where d_3 is a factor which depends on the size of the group and whose values have also been tabulated in standard statistical textbooks. Since σ is unknown, σ_R is estimated

by the following relationship:

$$\hat{\sigma}_R = d_3 \frac{\bar{R}}{d_2}. \quad (7)$$

Equations (5)–(7) form the basis for defining the three-sigma ($\alpha = 0.0027$) control limits for the $\bar{X}$ and the R charts:

$$CL_R = \bar{R} \pm 3d_3 \frac{\bar{R}}{d_2} \quad (8)$$

$$CL_{\bar{X}} = \bar{\bar{x}} \pm \frac{3}{d_2 \sqrt{n}} \bar{R} \quad (9)$$

where R is the average range out of a number (m) subgroups, each of size n , and $\bar{\bar{x}}$ is the *grand average*, also calculated from the same m subgroups. In Fig. 4, we give an example of applying an $\bar{X}$ - R chart for the control of the uniformity and the thickness of photoresist on silicon wafers.

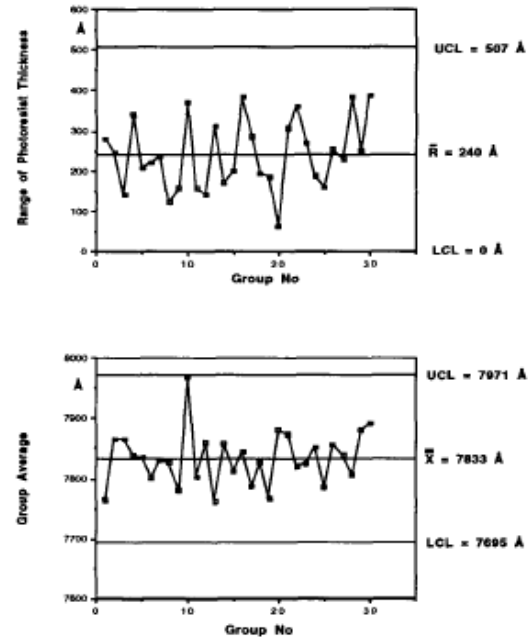


Fig. 4. $\bar{X}$ - R chart for the deposition of photoresist.

The R chart control limits for a given type I error are a function of the subgroup sample size, and can be derived from statistical tables. From these statistics we can also extract the operating characteristic function of the R chart, shown in Fig. 5, which is also adapted from [9]. Clearly, the R chart can detect with certainty only relatively large changes in the process spread.

2) *The $\bar{X}$ -S Charts*: When the subgroup size is small (less than about 5), the range of a group of measurements is a very efficient indicator of the process spread. When the subgroup size grows to more than 10, however, then the

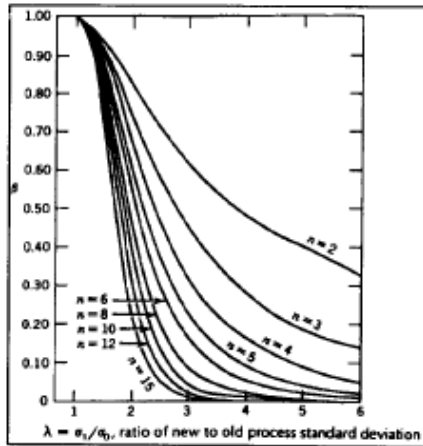


Fig. 5. Operating characteristic curves of the R chart.

familiar sum of squares is a better indicator of the process spread:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad (10)$$

This value, along with constants which depend on the group size, is used to derive the limits of the two companion charts. The derivation is similar to that of the $\bar{X}$ - R charts.

$$CL_s = \bar{s} \pm 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2} \quad (11)$$

$$CL_{\bar{x}} = \bar{x} \pm \frac{3s}{c_4 \sqrt{n}} \quad (12)$$

Here, c_4 is a proportionality factor which is needed to make σ an unbiased estimator of σ , according to the formula $E(s) = c_4 \sigma$. This factor depends on the group size n , and its value can be found in standard statistical tables, such as the ones in [9].

3) *Rational Subgrouping*: One note of caution must be inserted at this point: unless the variation that is affecting the group average (i.e., the group-to-group variation) is the same as the variation that determines the group spread (i.e., the within-the-group variation), the $\bar{X}$ limits should not be calculated from R . This means that (9) should not be used. This is a common error in semiconductor manufacturing applications, where the subgroup that looks most "natural" to the process engineer is the wafer. The causes and magnitude of the variation of a parameter *within* the wafer, however, are often very different than those of the variation of the same parameter *between* wafers. This is especially true in the case of single wafer processing, in which all points on a wafer experience a more or less uniform processing environment, although the run by run inconsistencies of the equipment often induce significant variation between wafers. Another problem might arise from deterministic radial patterns across a wafer. In this

case, (6) cannot be used to estimate the limits of the R chart. Pioneering work in processing these nonrandom spatial effects is discussed in [10].

4) *The Moving Range Chart*: Some parameters cannot be easily grouped, either because their readings are expensive, or because they are monitored continuously over a period of time. Temperature readings, for example, or readings resulting from expensive tests such as SEM measurements, cannot be easily grouped. Such parameters can be very effectively controlled by the moving range chart, a simple derivative of the $\bar{X}$ - R chart, where now the "group" is assumed to consist of two consecutive readings. In this way, group #1 consists of readings 1 and 2, group #2 includes readings 2 and 3, group #3 readings 3 and 4, etc.

The moving range chart is a powerful tool because of its simplicity. This chart permits the intuitive estimation of the in-control variance of the monitored parameter. As such, it is very useful for the control of continuously varying parameters, such as periodic temperature or pressure readings. One note of caution is necessary at this point: frequently sampled, continuously varying parameters are often *auto-correlated*, i.e., consecutive readings tend to depend on each other. Such parameters violate the IIND assumption (that readings should be *identically, independently, and normally* distributed). Even if the original parameter is IIND however, the consecutive differences will not be. The treatment of such parameters belongs to the rather advanced SPC chapter of time series analysis. The violation of the IIND assumption often results in false alarms through rules 6 and 7 of the Western Electric set as described in Section II-C.

As an example, Fig. 6 presents the application of the moving range chart for the control of real-time temperature readings from a polysilicon deposition furnace. In this example we are monitoring the temperature differential (i.e., the temperature reading at the center, minus the reading at the inlet of the reactor) via a moving range chart (Fig. 6(a)) and a chart of individual readings (Fig. 6(b)).

C. Runs Rules and the Western Electric Set

The control schemes discussed thus far deal with a very specific departure from the state of control—that of a clear shift in the value of the mean or the variance of the underlying distribution. In general, however, we are interested in seeing any sign of nonrandomness in our data. Indeed, nonrandomness may appear in many different ways, some of which do not involve a transgression of the three-sigma control limits. A popular set of runs rules is the *Western Electric* set, as summarized in Fig. 7.

Referring to part 6 of Fig. 7, consider the situation when we are recording the thickness of the photoresist layer. Although no three-sigma alarms are present, a large number of consecutive points appear to regularly alternate near the center line of the chart. This is a clear indication of nonrandomness that might result from some periodic maintenance pattern performed on the machine in question. Other such situations might lead to the stratification of

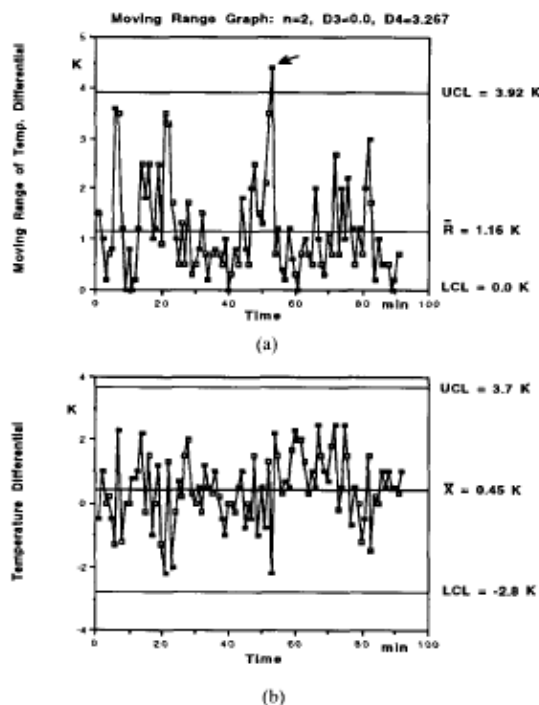


Fig. 6. A moving range chart for temperature control in a polysilicon deposition reactor. (a) The temperature differential via a moving range chart and (b) a chart of individual readings.

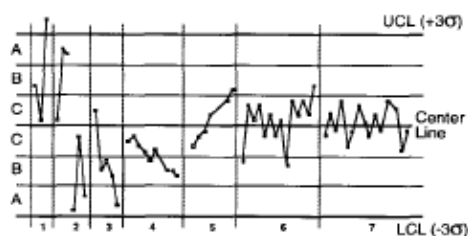


Fig. 7. Summary of the Western Electric rules.

- 1) Any point beyond three σ UCL or LCL.
- 2) 2/3 consecutive points on the same side, in zone A or beyond.
- 3) 4/5 consecutive points on the same side, in zone B or beyond.
- 4) 9/9 consecutive points on the same side of the centerline.
- 5) 6/6 consecutive points increasing or decreasing.
- 6) 14/14 consecutive points alternating up and down.
- 7) 15/15 consecutive points on either side in zone C.

points, where many consecutive points might appear in the same narrow region on the control chart.

So, in addition to the three-sigma limits, a number of "runs rules" have been introduced to identify such nonrandom situations. In general, the application of a runs rule involves the separation of the control chart into a number of zones. As an example, one such rule is used to issue an alarm whenever 4 out of 5 consecutive points fall between zero and three sigma on either the positive or the negative side of the center line. The violation of this rule is depicted in part 2 of Fig. 7.

In general, the application of multiple runs rules complicates the evaluation of the types I and II risks. A number

of complex simulators have been written to analyze the risks and the ARL of charts employing general sets of runs rules, and some are described in [12]. Runs rules have also been designed so that they optimize the cost effectiveness of a chart, taking into account the cost of each types I and II occurrence [9], [13]. Most of these schemes offer little intuition to the non-statistician, and they have not been used to a significant extent by the semiconductor manufacturing industry.

D. Controlling Defect Counts—An Example Using the Poisson Model

Many high complexity VLSI and ULSI products are vulnerable to defects that land on the wafer during processing. Consequently, ever since the inception of Integrated Circuits, a number of so-called *yield models* have appeared in the literature and have been used extensively by process engineers [14], [15]. The objective of these models is to predict the yield of a new IC design, given the defect density, the design rules, the die size, etc. Most of these models assume that the defect density is either constant, or that it obeys a known, stationary statistical distribution.

Once a new IC reaches production, however, it is usually accompanied by some modifications in the technology. This means that a new IC product usually starts at a low yield and it follows a "yield transient" while the process engineers learn the new process. Once an acceptable yield has been established, it must be monitored in order to ensure that the defect generation mechanism of the process remains under control. Another reason for statistical monitoring of the yield is to identify and quantify any yield changes that follow process modifications.

A distinct family of control charts, known as *attribute charts*, may be used in the control of process attributes such as the fraction of nonconforming die and the respective defect counts. These charts are based on statistical models that describe the particle generating mechanisms during processing. Although these types of charts are very simple, they directly monitor the fabrication line yield, which is, after all, one of the most important characteristics of a high volume production line.

The most direct method for monitoring yield is the direct application of the *fraction nonconforming* chart, also known as the *P chart*. In order to create such a chart we need to derive a relevant statistical model of the production process. This model is based on the assumptions that: a) the process is operating without any assignable causes, and b) each die has a constant probability p of being defective.

Under these assumptions, if we sample n die at a time, the probability that we will find x defectives ($P\{D = x\}$) is given by the binomial distribution:

$$P\{D = x\} = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 1, 2, \dots, n. \quad (13)$$

If we measure the proportion of defective die from multiple groups (lots) and use the average as the monitored statistic, the mean is equal to the probability of failure p and the variance is also known. More specifically, if we count the

defective die (D) out of a group of n die, and if we use m of these groups to establish the control chart, then the centerline of the chart is given by

$$\bar{p} = \frac{\sum_{i=1}^m \hat{p}_i}{m} \quad (14)$$

where the estimated fraction nonconforming for each group is given by

$$\hat{p}_j = \frac{D_j}{n}, \quad j = 1, 2, \dots, m. \quad (15)$$

Finally, the three-sigma control limits are given by (16) below, assuming that the sample size n is large enough so that the binomial distribution is almost symmetrical about its mean. This implies that it can be approximated by a normal distribution. In this case the control limits are given by

$$CL = \bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}. \quad (16)$$

There are several rules which deal with the assumption of symmetry and with the design of the P chart in general. These rules are described in detail in [9].

Similar charts can be used to monitor the number of defects when assuming a known, constant defect density c and a mechanism that generates defects according to a Poisson distribution. This control chart is known as the C chart, and its three-sigma control limits are given around the known defect density c . This defect density represents the average number of defects on each inspection unit, which might be a die, a wafer, or a batch of wafers.

$$CL = c \pm 3\sqrt{c}. \quad (17)$$

Another useful attribute chart is the U chart, which deals with the average defect count over a group of n entities such as die, wafers, or wafer batches. The control limits of the U chart are based on averaging the Poisson-distributed defect counts. Thanks to the Central Limit Theorem, this average will tend to be distributed according to a Gaussian distribution. Therefore, the three-sigma limits of the U chart are given by

$$CL = \bar{u} \pm 3\sqrt{\frac{\bar{u}}{n}} \quad (18)$$

where $\bar{u}$ is the observed average defect density over n inspection units.

The P , C , and U charts have traditionally been based on a model that describes the random generation of defects according to a Poisson distribution. The Poisson-based model, however, is unable to describe the clustering effect which causes defects to appear in groups on some of the larger IC products. Recently, more sophisticated models have been proposed to extend the application of formal control schemes to cases that cannot be modeled with a Poisson distribution [16].

The P chart example that follows in Fig. 8(a) uses data from [15], and monitors the number of defective die on

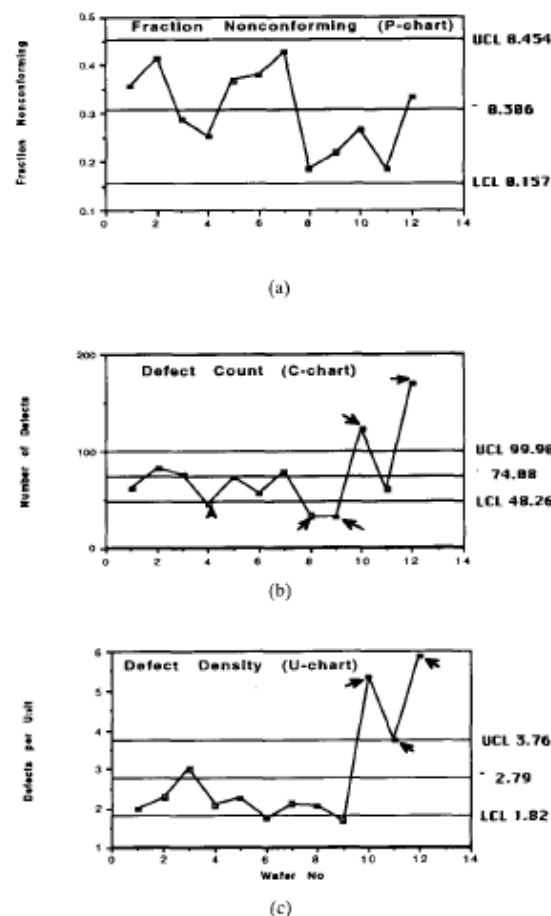


Fig. 8. (a) The P chart, built to monitor the portion of bad die per wafer. (b) The C chart, used to monitor defects/wafer. Limits are based on the Poisson Model. (c) The U chart, used to monitor the average number of defects per die on each wafer.

each wafer, one wafer at a time. Two additional charts can be applied here to help focus not so much on the IC product (wafer) but on the *causes* behind the yield fluctuation. The C chart in Fig. 8(b) shows the number of defects on each wafer, assuming that the defects are generated according to an unclustered Poisson distribution. The U chart in Fig. 8(c) monitors the average number of defects/defective die for each wafer. It is also based on unclustered Poisson statistics. Although these tools cannot account for clustering, they are powerful, straightforward tools that can successfully detect yield fluctuations.

E. Maximum Likelihood Estimation Control—The CUSUM Chart

A newer class of control charts, introduced in the late 1950's, is based on the concept of *maximum likelihood*. These charts use the cumulative sum (CUSUM) of process deviations in order to generate an alarm [17]. The approach is quite sensitive to small, persistent deviations of a process, such as those due to subtle miscalibrations or small changes in the quality of incoming material. Since most semiconduc-

tor processes are well instrumented against large deviations, CUSUM schemes can effectively capture the remaining small deviations and are very suitable to semiconductor process control. Here, the monitored statistic is equivalent to the accumulated deviation of the recorded parameter from its target:

$$\sum_{i=1}^n (\bar{u} - \mu_0). \quad (19)$$

The formulas necessary to produce a chart based on this statistic are given below:

$$d = \left(\frac{2}{\delta^2} \right) \ln \left(\frac{1 - \beta}{\alpha} \right) \quad (20)$$

$$\theta = \arctan \left(\frac{\Delta}{2A} \right) \quad (21)$$

$$\delta = \frac{\Delta}{\sigma_{\bar{x}}} \quad (22)$$

$$d = -\frac{2 \ln \alpha}{\delta^2}. \quad (23)$$

Here, d is the lead distance (in number of samples) and θ is the angle of the V-shaped limits. The type I error of this chart is α . δ is defined as the deviation to be detected with a type II error β . The same deviation, expressed in number of sigmas of the sampling average, is δ . Finally, a scaling factor is needed to relate the vertical to the horizontal scales in the graph, so that the angle of the V-shaped limits is correctly drawn. This scaling factor is A and it is usually given values between 1 and 2 s, where s is the estimated standard deviation of x . An example of the application of the CUSUM chart is shown in Fig. 9.

Due to the inherent smoothing of the CUSUM chart, (the integration acts as a low-pass filter that effectively eliminates any spikes), this scheme is ideal for automatic feedback control applications.³ This way meaningful long term changes can be observed and compensated separately from unique disturbances.

Another example of the application of the CUSUM chart in semiconductor manufacturing is drawn from the run by run control of a photolithographic workcell. In this application we are interested in maintaining consistent levels of photoactive compound concentration within the photoresist layer. The level of concentration can be inferred by means of a specialized reflectance measurement [19]. Any change in the measured reflectance can lead to assignable causes of variation in the consistency of our photoresist supply or in the consistency of operation of our spin/coat and bake equipment. As can be seen in the accompanying figures, the CUSUM chart in Fig. 10 responds faster and gives a

³ Another chart with inherent smoothing capabilities is the exponentially weighted moving average control scheme (EWMA) [18]. Although not yet popular in semiconductor manufacturing, this scheme offers a reasonable compromise between the large shift responsiveness of the Shewhart chart and the small shift sensitivity of the CUSUM chart.

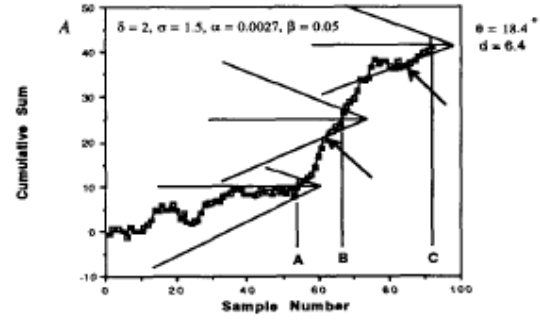


Fig. 9. CUSUM chart for temperature control during poly deposition.

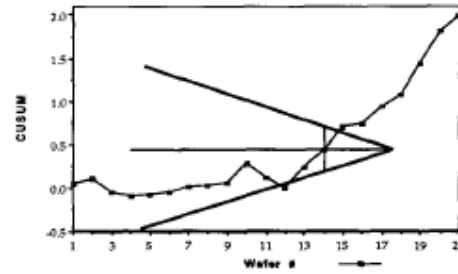


Fig. 10. CUSUM chart of measured photoresist reflectance.

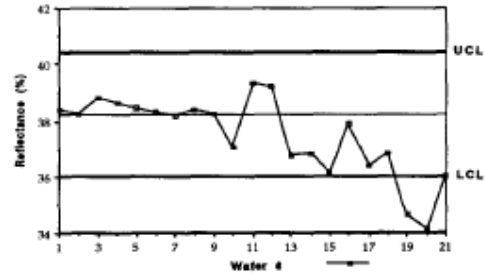


Fig. 11. Shewhart chart of the photoresist reflectance pictured in Fig. 10.

more unambiguous picture of the Shewhart chart in Fig. 11. In this case, the assignable cause has been found to be a miscalibration of the prebake temperature.

III. NOVEL SPC METHODS IN SEMICONDUCTOR MANUFACTURING

Although SPC has been applied on high volume production since the early 1930's, the original techniques have evolved significantly over the years in order to accommodate the needs of changing manufacturing technology. A major force behind the evolution of statistical process control is the recent availability of automated *in situ* data collection and real-time data processing capabilities. As a result, comprehensive control schemes which would have been impractical two decades ago are now finding their way

onto the factory floor. This revolution is likely to have a major impact on semiconductor manufacturing.

The special control requirements of semiconductor manufacturing stem from the poor repeatability of several critical VLSI manufacturing steps and also from the need to achieve high *process capability*⁴ with technologies that have little time to mature before they are applied in production. This situation becomes even more complicated when typical production runs between recipe changes are short. In addition, in order to achieve high reliability of production, several critical processes should be monitored by means of multiple real-time parameters. Unfortunately, these parameters are typically cross-correlated and non-IIND.⁵

These circumstances create special requirements and opportunities for the application of SPC in semiconductor manufacturing. In the rest of this paper we will describe some of these special techniques, including multivariate, model-based and real-time applications of statistical process control in semiconductor manufacturing.

A. Multivariate Control—Hotellings T^2 Chart

Often, a critical processing step might be monitored by means of recording several parameters. One example of this is the monitoring of dry polysilicon etching through the etch rate, etch uniformity, selectivity to photoresist and selectivity to oxide. Although these measurements carry important information about the process, they also need special SPC schemes for their analysis. More specifically, an important consideration is the fact that such parameters are very likely to be statistically correlated with each other. This means that if we use a number of independent control charts, the overall manufacturer's (α) and consumer's (β) risks cannot be evaluated correctly.

In response to this problem, several multivariate control techniques have emerged and are in use today. These schemes alert the operator to changes in the mean vector or the covariance matrix of a group of controlled parameters. One of the most popular multivariate control schemes is based on Hotelling's T^2 statistic. This statistic, defined below, is sensitive to the collective deviations of a number of cross-correlated IIND parameters from their respective targets. Assuming that we have p such parameters whose variance-covariance matrix is known and does not change (even if the process goes out of control), the T^2 statistic is given by the formula:

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^T \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \quad (24)$$

where n is the size of each measurement subgroup, $\bar{\mathbf{x}}$ is the vector of the group averages as measured, $\boldsymbol{\mu}$ is the vector

⁴The *process capability* measures of C_p (used when a process is centered around its specifications) and C_{pk} (for skewed processes) are related to how suitable is a process for the application at hand. C_p is defined as the ratio of the specification window over the six-sigma spread of the process, while C_{pk} is similarly defined for potentially skewed processes.

⁵Identically, independently, and normally distributed. See Section II-A.

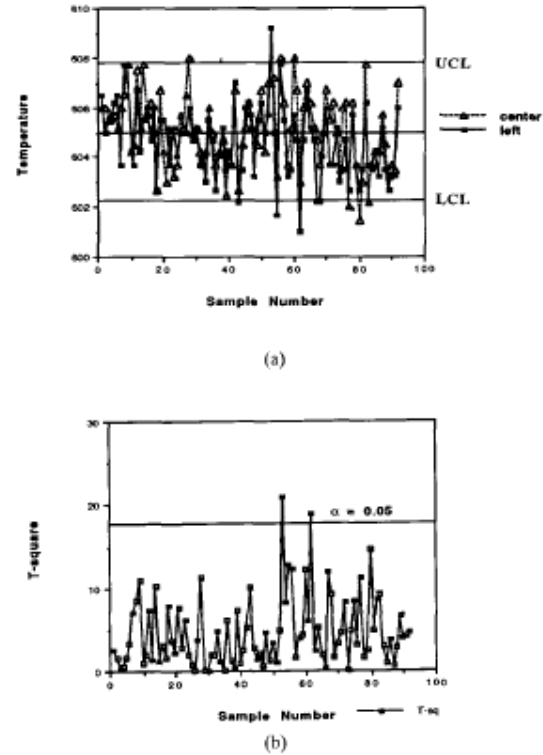


Fig. 12. (a) Center and left temperature averages (4 readings per group) in LPCVD furnace. (b) T^2 plot for center and left temperature average.

of the group means (target values), and $\mathbf{S}$ is the $p \times p$ covariance matrix. The superscript (T) is used to indicate the transpose operation. All the vectors are originally defined as $1 \times p$ arrays (i.e., columns). Under the assumption that, when under control, all the random variables are identically and independently, normally distributed (IIND) around their respective mean μ_i , then the α -level upper control limit of this one-sided chart is given with the help of the chi square distribution:

$$UCL = \chi_{\alpha, p}^2 \quad (25)$$

If the parameter mean and the covariance matrix have been estimated from a small number of samples, then the upper control limit is more correctly defined with the help of the F distribution [26].

An example of the T^2 statistic is shown in Fig. 12(a) and (b). Here we use two temperature readings at either end of an LPCVD deposition tube in order to monitor the temperature during the deposition of critical polysilicon films. In Fig. 12(a) we present two control charts with limits set for $\alpha = 0.05$. Since the process was under control, we would only expect to see about five false alarms, yet we see many more. In Fig. 12(b) the one-sided control limit has also been set for $\alpha = 0.05$, but now we only receive two false alarms. Obviously, the T^2 statistic presents a far clearer picture of the process status and is much less likely to introduce false alarms.

B. Model-Based Statistical Process Control—The Regression Chart

Semiconductor manufacturing equipment is often used to support an array of products, each requiring a different processing recipe. According to traditional SPC practice however, multiple recipes (or intentional changes of any kind) cannot be present if a control chart is to be used. This requirement is very restrictive in semiconductor manufacturing, where change is the rule and long runs are the exception. A variant of SPC, known as *model-based SPC* can be used to solve this problem.

The foundation of model-based SPC is the Regression Chart [21], introduced by Mandel in 1969.⁶ Instrumental in the regression chart is the use of a regression model that predicts the nominal response of the various equipment as a function of their settings. The *residual* of this response is obtained as the difference between the predicted and the observed equipment response. Since the statistics of the residual are well known, a Shewhart control chart can be used to control it. Out of control points are then treated as indications of assignable causes.

An important special application of the regression chart concerns points that appear to be *systematically* out of control. This usually means that the equipment has drifted and that the models must be reevaluated. This situation can be detected with the help of the cumulative student-*t* statistic. In this way, an *adaptive* statistical process control scheme can be implemented so that abrupt changes can be detected, even as the equipment model is being continuously adapted. This model can then be used for further process optimization.

An example of such an application appears in Fig. 13(a), where the low-pressure chemical vapor deposition tube is being controlled by a regression chart. The regression equation has been built to predict the deposition rate as a function of temperature, SiH_4 flow and pressure [12], [24]. The control limits reflect both the experimental error of the tube as well as the prediction error of the regression model. The situation depicted in Fig. 13 shows that the tube is out of control, since several of the plotted points violate the control limits. In the first graph in Fig. 13 the regression model is also significantly different than the equipment response. In the second graph the model has been *adapted* (recentered), yet this adaptation did not interfere with the primary detection of one of a kind assignable causes. Similar schemes are now in use for the control of experimental photolithographic operations in the Berkeley Microfabrication Laboratory. The detailed description of the model-based SPC scheme will be the subject of a future publication.

C. Time Series Analysis

Modern semiconductor manufacturing equipment are outfitted with sensors capable of monitoring a number of

⁶The term *model-based SPC* has recently been reintroduced in [25] to describe a similar concept where the model residuals are applied on a Shewhart chart. This simplified scheme does not take into account the prediction error of the regression equation.

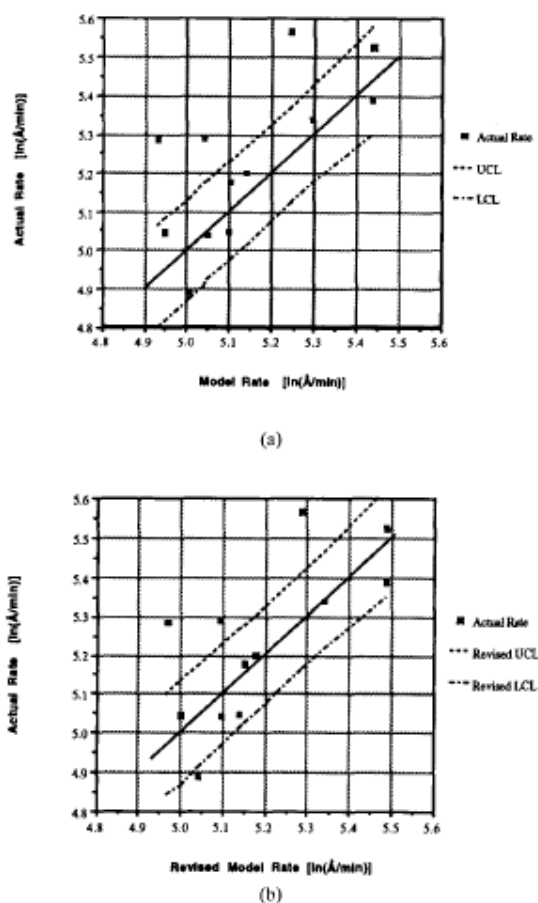


Fig. 13. (a) Model-based SPC on an LPCVD tube. Systematic error indicates chronic drift. (b) Model-based SPC on LPCVD tube. Model has been adapted to account for chronic drift.

critical process parameters such as temperature, pressure, gas flows etc. In addition, most new equipment can automatically upload these readings to a host computer system with the help of the SECSII [20], the standard interequipment communication protocol instituted by SEMI.

A problem that often arises in conjunction with such rapid, continuous parameter readings is that each new value tends to be *statistically related* to previously measured values. The existence of autocorrelation in the controlled parameters violates one of the most basic assumptions behind the design of standard SPC schemes, namely that all samples are *IIND* random variables. In order to cope with this problem, the monitored parameter might be modeled by means of an appropriate *time series* model. Time series models, such as the well known autoregressive integrated moving average (ARIMA), can be used to forecast each measurement and deduce the *forecasting error* [22]. This error can then be assumed to be an independently distributed random variable, and it can be used with traditional SPC schemes. An example of this application appears in Fig. 14. In this example, we examine the real-time readings

of temperature collected from a low pressure, chemical vapor deposition tube. In Fig. 14(b) we see significant autocorrelation between temperature readings separated by one time period. Based on this autocorrelation, a model is generated to *forecast* each new reading. Finally, in Fig. 14(c) the forecasting error is treated as an IIND parameter. Compare Fig. 14(a) to Fig. 14(c) and notice the appreciable reduction of alarms. The limits are set for $\alpha = 0.05$. The model used in this simple example is the ARIMA (1,0,0) model since it involves one autoregressive term, no differentiation and no moving average terms.

The power of time series models becomes apparent when one considers the recent popularity of multichamber equipment [27]. These so-called "cluster" tools promise much improved process quality by automatically sequencing a wafer through several processing steps without exposing it to the cleanroom atmosphere during transport. This, however, makes it impossible for the operator to inspect the wafers between steps in order to make sure that individual operations remain under control. The only way to achieve this is through the statistical monitoring of real time sensor readings, and this is only possible with the help of the appropriate time series filtering [28].

IV. COMPUTER-INTEGRATED MANUFACTURING AND SPC

Easy access to information, both for generating an alarm as well as for discovering its assignable cause, is instrumental in SPC operations. With current computer technologies, it is possible to construct a physically distributed but logically integrated database. This will greatly facilitate data manipulation across the manufacturing floor and will lead to high productivity.

Four recent advances contribute to this realization. The most important advance is the development of relational database systems which reduce the effort required for both the initial development and the subsequent maintenance and modification of a system. This is because relational databases support interfaces which allow end-users (in this case process, maintenance, and yield engineers) to easily manipulate the information stored in the database.

The second major development has been the industry-wide acceptance of high bandwidth communication standards (local area networks, or LAN's) for linking systems from different vendors. LAN's make it possible to connect process control applications directly to the fabrication equipment. Consequently, in-line and in-process measurements can be automatically collected and analyzed.

A third important development is the emergence of distributed database management systems. Due to distributed database systems, information is physically stored at many nodes yet appears to the user as a coherent entity. The distributed database system determines where the data is located, generates an efficient plan to retrieve or update it, and ensures its consistency and integrity.

The fourth important element is the spreading use of artificial intelligence technologies. Many knowledge intensive, error-prone activities in semiconductor manufacturing

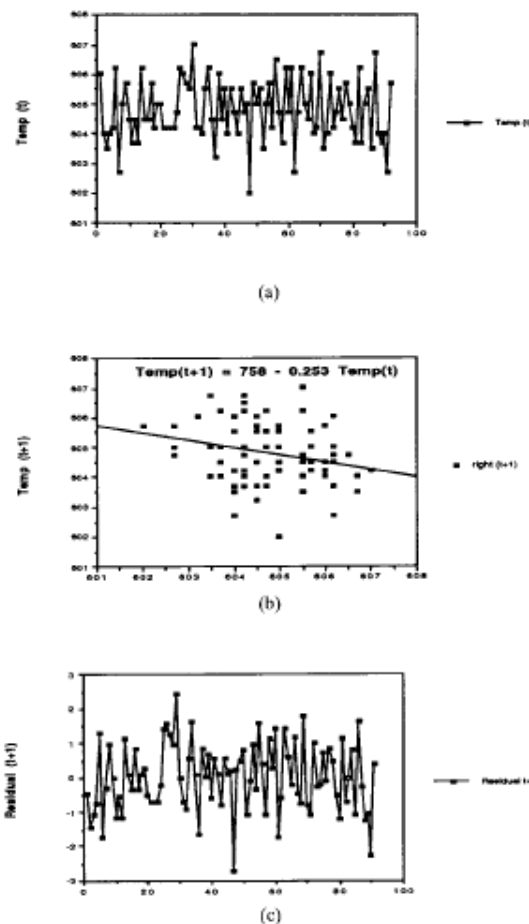


Fig. 14. (a) Filtering real-time non-IIND data for SPC. "Raw" temperature readings over time. $\alpha = 0.0027$. (b) Filtering real-time non-IIND data for SPC. Autocorrelation in temperature readings. (c) Filtering real-time non-IIND data for SPC. Chart of IIND residuals from ARIMA (1, 0, 0).

can be automated by the use of AI techniques. The need for automated decisions in planning, scheduling, diagnosis, maintenance, etc. becomes even more pressing in view of the complexities of the new submicron ULSI processes. The objective of the Berkeley CIM architecture is to develop software modules for controlling VLSI processing steps, and to demonstrate a flexible framework for combining these modules into an integrated CIM system.

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